Electric Vehicle Market Performance & Insights

FINAL PROJECT REPORT

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Prepared according to the uploaded Project Report Format.

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1. INTRODUCTION

1.1 Project Overview

This project analyses the supplied electric-vehicle dataset, ev_market_test.csv, using Tableau Public. It presents market-performance and technology-oriented views covering manufacturers, recorded annual sales, battery technologies, driving range, battery capacity, manufacturing-country records, customer ratings and model years.

The project deliverables comprise one interactive Tableau dashboard, eight visualisations, two Tableau stories and a public project website hosted through GitHub Pages.

1.2 Purpose

The purpose is to organise the supplied data into clear visual insights that help a viewer explore EV-related patterns and compare manufacturers and technologies. The dashboard is intended for market-strategy, business-intelligence and sustainable-mobility research audiences.

2. IDEATION PHASE

2.1 Problem Statement

EV-related information can be difficult to compare when measures such as sales, range, battery capacity, technology type and customer ratings are viewed separately. The project addresses this analysis and communication problem by bringing the available fields into a single visual analytics experience.

Scope note: the project uses a supplied test dataset. It does not establish verified global EV market conditions.

2.2 Empathy Map Canvas

The project was framed around three stakeholder perspectives: an EV market strategy director interested in manufacturer and category performance; an automotive business-intelligence manager interested in comparisons and market indicators; and a sustainable-mobility research analyst interested in battery technology, range and manufacturing distribution.

These personas are design perspectives used to guide the analysis, not findings from documented customer interviews.

2.3 Brainstorming

Candidate analysis ideas included manufacturer sales ranking, battery technology distribution, average driving range by manufacturer, battery capacity versus range, manufacturing-country distribution, customer satisfaction, battery capacity by technology and annual sales grouped by model year. The final project uses these eight views.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map

A typical viewer journey is: open the project website; read the project overview and methodology; open the interactive Tableau dashboard; select available filters; compare manufacturers, technologies and performance measures; then explore the two Tableau stories for a guided narrative.

3.2 Solution Requirement

Functional requirements: connect the supplied CSV; inspect data quality; create eight visualisations; assemble an interactive dashboard; publish two stories; and provide public access through a website with direct Tableau links.

Non-functional requirements: clear labels, consistent visual design, accessible navigation, readable charts, transparent methodology and explicit communication of dataset limitations.

3.3 Data Flow Diagram

Data flow: supplied ev_market_test.csv → data inspection and preparation → Tableau Public data connection and aggregations → eight visualisations → interactive dashboard and two stories → GitHub Pages website → project viewers.

No Snowflake database, warehouse, SQL pipeline or ETL service is part of the implemented solution.

CSV dataset	Data checks	Tableau Public	Dashboard + Stories	GitHub Pages

3.4 Technology Stack

Data: CSV file ev_market_test.csv. Analysis and visualisation: Tableau Public. Website: HTML and CSS (with the supplied static website files). Hosting and source repository: GitHub Pages and GitHub. Python/pandas was used for dataset profiling and preparation checks documented in the project materials.

4. PROJECT DESIGN

4.1 Problem Solution Fit

The selected solution matches the project problem by consolidating related EV measures into a dashboard and two guided stories. It enables side-by-side exploration using charts and available dashboard filters instead of requiring users to inspect raw rows.

4.2 Proposed Solution

Build a public-facing analytics experience consisting of one dashboard and two stories. Story 1, EV Market Performance Overview, focuses on manufacturer performance, recorded sales, battery-technology distribution and manufacturing-country records. Story 2, EV Technology and Customer Insights, focuses on range, capacity, battery technology and customer ratings.

4.3 Solution Architecture

The CSV is the source layer. Data quality checks are applied before analysis. Tableau Public provides the analysis and visualisation layer. The dashboard and stories are published on Tableau Public and linked or embedded from the GitHub Pages website. The website acts as the public project landing page.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Work was organised into stages: ideation and stakeholder framing; requirements and solution design; dataset inspection and preparation; Tableau visualisation and dashboard development; story creation; testing and review; website integration; and final documentation.

Earlier planning materials proposed four six-day sprints and 80 story points as a planning model. These are planning estimates, not verified actual team velocity or completion measurements. Team member is recorded as Self; Team ID is intentionally left blank.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Dataset profile recorded in the project work: 34,259 rows and 37 columns; 34,259 distinct vehicle_id values; model_year values from 2018 to 2026; and zero exact duplicate rows. Five columns contain missing values: battery_cycle_life (691), resale_value_pct_3yr (1,026), infotainment_screen_inch (529), annual_sales_units (362) and customer_rating (690), totalling 3,298 missing cells.

Missing values were retained as unknown rather than replaced with zero. Repeated vehicle IDs were not found. Text whitespace and negative values in selected numeric measures were checked. The supplied file is a test dataset, so results should be interpreted within its scope.

Tableau functional checks should confirm that the dashboard opens, all eight views render, available filters update the relevant views, and both published stories open. No measured response-time benchmark or Snowflake query execution time was supplied; therefore no numeric latency or query-performance claim is made.

7. RESULTS

7.1 Output Screenshots

The project website and published Tableau dashboard are the intended output evidence. Insert current screenshots captured directly from the live website and Tableau Public pages in this section before final academic submission. The report does not substitute generated chart previews or invented screenshots for live application evidence.

Published dashboard and story links are provided in the Appendix for direct review.

[Insert Screenshot 1: Public project website home page]

[Insert Screenshot 2: Tableau dashboard showing all eight visualisations]

[Insert Screenshot 3: Tableau Story 1 and Story 2]

8. ADVANTAGES & DISADVANTAGES

Advantages

• Combines eight EV analysis views in one dashboard.
• Provides two guided Tableau stories for different analytical themes.
• Offers public access through a GitHub Pages website.
• Makes it easier to compare manufacturers, technologies and selected performance measures.
• Documents missing values and interpretation limits.

Disadvantages / Limitations

• Findings depend on the supplied test dataset and are not verified global market statistics.
• Missing values affect the available sample for some averages and totals.
• Country record counts do not measure national factory output.
• Battery-type record counts are not real-world market shares.
• Model-year sales are grouped by model_year, not by transaction date.
• No Snowflake backend or measured query-latency benchmark is implemented.

9. CONCLUSION

Conclusion

The project delivers a public EV analytics solution using a CSV dataset, Tableau Public and GitHub Pages. It includes eight visualisations, one interactive dashboard and two published stories. The result provides a structured way to explore the supplied dataset while making its scope and limitations explicit.

10. FUTURE SCOPE

Future Scope

Future improvements could include obtaining a validated real-world EV dataset; refreshing the data regularly; adding verified time-series and regional comparisons; improving mobile dashboard layout; adding documented performance benchmarks; collecting feedback from actual users; and extending the analysis with additional battery, charging, price and adoption indicators where reliable data is available.

11. APPENDIX

Source Code (if any)

The public GitHub repository is the project source-code and website reference. The implementation is primarily a Tableau workbook and static website; no separate Snowflake SQL source code is claimed.

Dataset Link

Dataset used: ev_market_test.csv (supplied project dataset). A public dataset download URL was not provided in the project details. If the CSV is committed to the GitHub repository, cite the repository path or direct file URL here; otherwise attach the CSV separately to the submission.

GitHub & Project Demo Link

GitHub repository: https://github.com/suryadevsingh0225/Electric-Vehicle-Market-Performance-Insights

GitHub Pages website: https://suryadevsingh0225.github.io/Electric-Vehicle-Market-Performance-Insights/

Tableau dashboard: https://public.tableau.com/app/profile/suryadev.singh/viz/EV_Market_Performance_Insights_17931646155280/ELECTRICVEHICLEMARKETPERFORMANCEINSIGHT?publish=yes

Story 1 – EV Market Performance Overview: https://public.tableau.com/app/profile/suryadev.singh/viz/EV_Market_Performance_InsightsStory1/EVMarketPerformanceOverview?publish=yes

Story 2 – EV Technology and Customer Insights: https://public.tableau.com/app/profile/suryadev.singh/viz/EV_Market_Performance_InsightsStory2/EVTechnologyandCustomerInsights?publish=yes

Demo video: not provided; add the recording URL if the submission requires a video.

https://github.com/suryadevsingh0225/Electric-Vehicle-Market-Performance-Insights

https://suryadevsingh0225.github.io/Electric-Vehicle-Market-Performance-Insights/

https://public.tableau.com/app/profile/suryadev.singh/viz/EV_Market_Performance_Insights_17931646155280/ELECTRICVEHICLEMARKETPERFORMANCEINSIGHT?publish=yes

https://public.tableau.com/app/profile/suryadev.singh/viz/EV_Market_Performance_InsightsStory1/EVMarketPerformanceOverview?publish=yes

https://public.tableau.com/app/profile/suryadev.singh/viz/EV_Market_Performance_InsightsStory2/EVTechnologyandCustomerInsights?publish=yes

Submission checklist: Add live output screenshots, confirm the dataset is included or attached, and add a demo video URL if required by the course portal.